Date of announcement: 28/10/22 Date of Submission: 2/11/22

Assignment - 2

6/6

5. Explain the N-Queen's problem and its solution using Backtracking.(6 Marks)

- N-Queens Problem is to Place n-queens in such a manner on an $n \times n$ chessboard that no queens attack each other by being in the same row, column or diagonal for $n=1$, the problem has a trivial solution and no solution exists for $n=2$ and $n=3$

N = 2



N = 3

Place 3 queens in 3x3 chessboard.

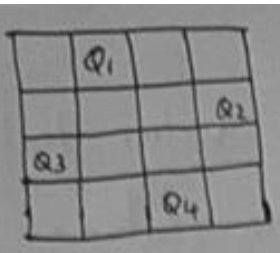
		q_1
q_2	$q_2 x$	$q_2 x$
$q_3 x$	$q_3 x$	$q_3 x$

No solution

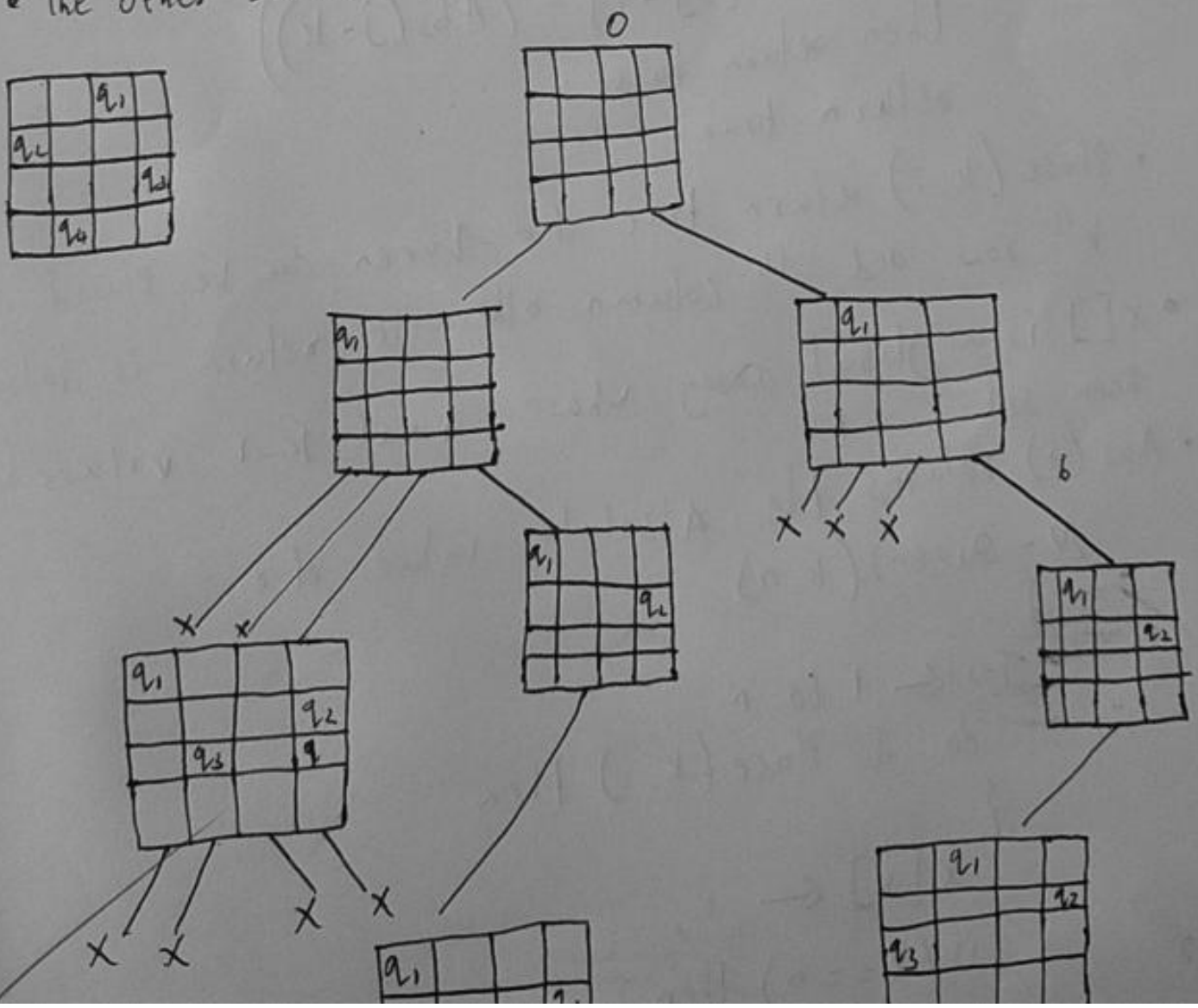
- first we will consider the 4-queens problem and then generate it to n-queens Problem. Given a 4x4 chessboard and number the rows and column of the chessboard 1 through 4.
- Since we have to Place 4 queens such as q_1, q_2, q_3 and q_4 on the chessboard, such that no two queens attack each other. Each queen must be placed on a different row ie, we put queen's "i" on row "i"

Solution

- we place queen q_i in the very first acceptable position $(1,1)$
- Next, we put queen q_2 so that both these queens do no attack each other
- we find that if we place q_2 in column 1 and 2, then the dead end is encountered.



- Now we backtrack one step and place the queen 'q_2' in (2,4) the next best Possible solution
- Then we obtain the Position For Placing 'q_3' which is (3,2)
- But later this Position also leads to a dead end, and no Place is found where 'q_4' can be Placed Safely.
- Then we have to backtrack till 'q_1' and Place it to (1,2) and then all other queens are Placed safely by moving q_2 to (2,4) q_3 to (3,1) and q_4 to (4,3)
- That is we get the solution (2,4,1,3). This is one Possible solution For the 4 queen Problem
- for another Possible solution the whole method is repeated for all Partial solutions.
- The other solutions for 4 queens Problem is (3,1,4,2)



Algorithm

- Place (k, i) returns a boolean that is true if the kth queen can be placed in column i.
- It tests both whether i is distinct from all previous costs $x_1, x_2 \ldots x_{k-1}$ and whether there is no other queen on the same diagonal
- Using Place, we give a precise solution to the n-queens Problem:

```
Place (k, i)
{
  for j ← 1 to k-1
  do if (x[j] = i)
   or (Abs x[j] - i) = (Abs (j - k))
  then return false;
  return true;
```

- Place (k, i) return true if a queen can be placed in the kth row and i^{th} column otherwise return is false
- x[] is a global array whose final k-1 values have been set
- Abs (r) returns the absolute value of r

```
N-Queens (k, n)
{
 for i ← 1 to n
   do if Place (k, i) then
   {
     x[k] ← i;
     if (k == n) then
     write (x[1 ... n]);
     else
     N-Queens (k+1, n);
}}
```

3/11/2022